Active Record Basics

1 What is Active Record?

Active Record is the M in MVC - the model - which is the layer of the system responsible for representing business data and logic. Active Record facilitates the creation and use of business objects whose data requires persistent storage to a database. It is an implementation of the Active Record pattern which itself is a description of an Object Relational Mapping system.

1.1 The Active Record Pattern

Active Record was described by Martin Fowler in his book *Patterns of Enterprise Application Architecture*. In Active Record, objects carry both persistent data and behavior which operates on that data. Active Record takes the opinion that ensuring data access logic as part of the object will educate users of that object on how to write to and read from the database.

1.2 Object Relational Mapping

Object Relational Mapping, commonly referred to as its abbreviation ORM, is a technique that connects the rich objects of an application to tables in a relational database management system. Using ORM, the properties and relationships of the objects in an application can be easily stored and retrieved from a database without writing SQL statements directly and with less overall database access code.

1.3 Active Record as an ORM Framework

Active Record gives us several mechanisms, the most important being the ability to:

- Represent models and their data.
- Represent associations between these models.
- Represent inheritance hierarchies through related models.
- Validate models before they get persisted to the database.
- Perform database operations in an object-oriented fashion.

2 Convention over Configuration in Active Record

When writing applications using other programming languages or frameworks, it may be necessary to write a lot of configuration code. This is particularly true for ORM frameworks in general. However, if you follow the conventions adopted by Rails, you'll need to write very little configuration (in some cases no configuration at all) when creating Active Record models. The idea is that if you configure your applications in the very same way most of the time then this should be the default way. Thus, explicit configuration would be needed only in those cases where you can't follow the standard convention.

2.1 Naming Conventions

By default, Active Record uses some naming conventions to find out how the mapping between models and database tables should be created. Rails will pluralize your class names to find the respective database table. So, for a class `Book`, you should have a database table called **books**. The Rails pluralization mechanisms are very powerful, being capable to pluralize (and singularize) both regular and